

Seen the right way, a page would let the agent solve up to 16 more tasks in 100 than the best fixed mode — yet none of 8 learned routers beat always using the cheapest mode on both success and cost.

WHERE THE DECISION SITS IN THE AGENT LOOP

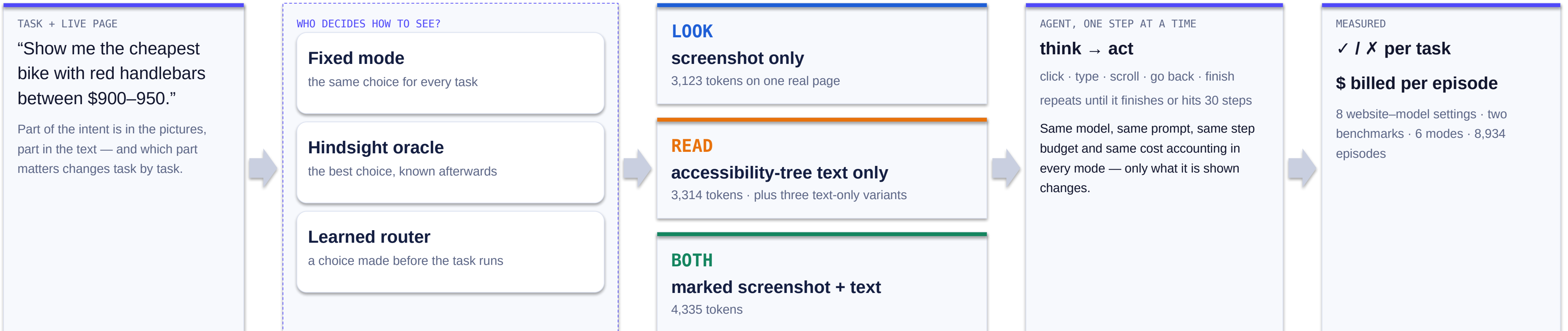


Fig 1. Everything is held fixed except how the page is shown. The dashed box is what this work measures: a fixed choice, the best choice in hindsight, and a choice a router has to learn — each judged on **both** success and cost. The laptop beside this poster replays the three eyes on the three tasks below.

ON THE SCREEN BESIDE YOU

“Navigate to the item on this page whose image is taken during a sunset.”

LOOK ✓ 2 steps \$0.007

READ ✗ 21 steps \$0.096

BOTH ✓ 3 steps \$0.014

“Navigate to my listing of the blue bike and change the price to \$85.50 (including in the description).”

LOOK ✗ 30 steps \$0.116

READ ✓ 9 steps \$0.079

BOTH ✗ 6 steps \$0.029

“Show me the cheapest bike with red handlebars between \$900-950.”

LOOK ✗ 19 steps \$0.075

READ ✗ 6 steps \$0.025

BOTH ✓ 7 steps \$0.037

One recorded run per mode, B0 · classifieds. Every ✓ / ✗ above came out the same on an independent rerun; steps and cost differ run to run — across all tasks a rerun flips 10–14% of outcomes.

RESULTS ACROSS 8,934 EPISODES

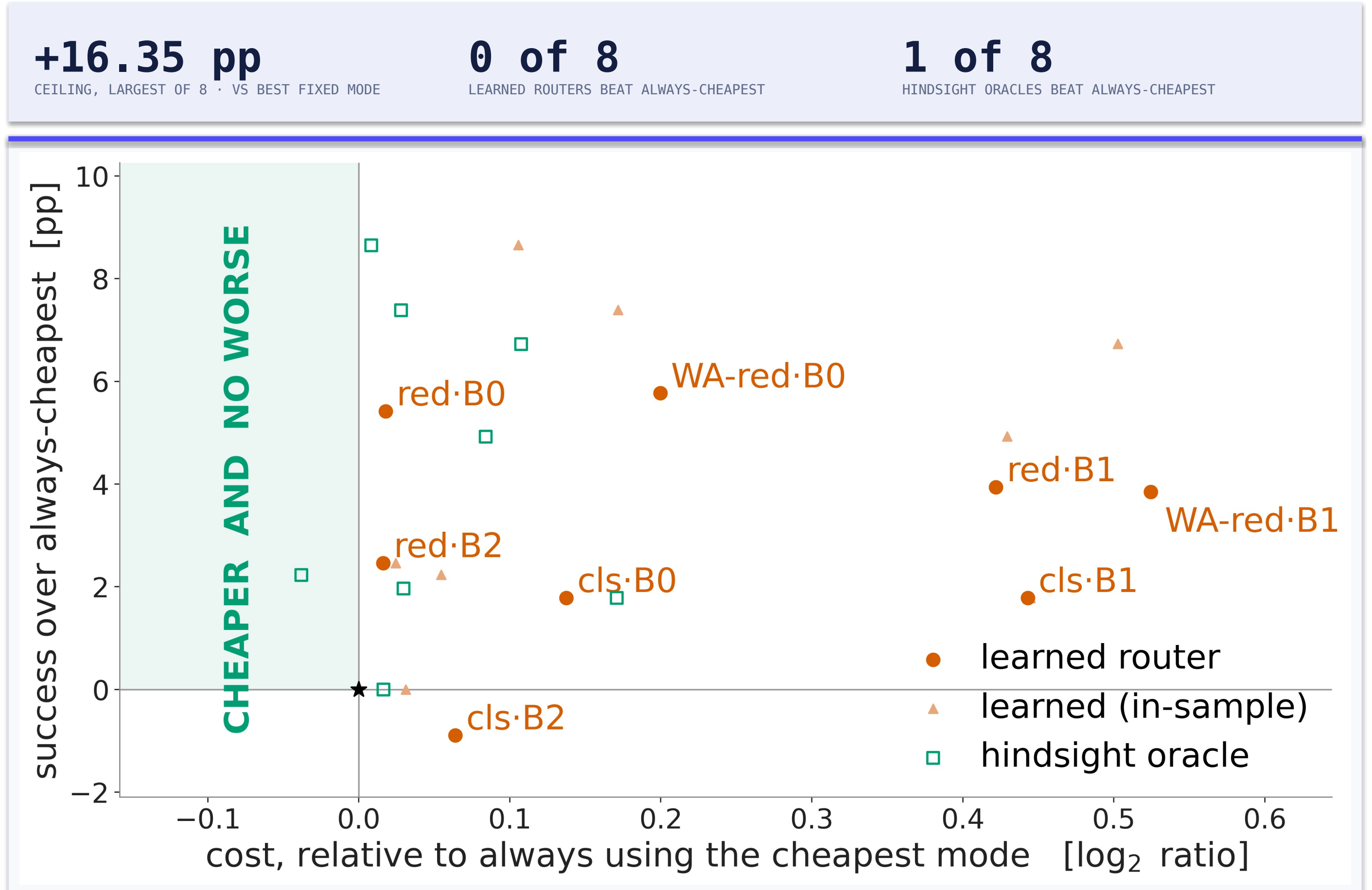


Fig 2. Every policy in every setting against one fixed baseline, **always use the cheapest mode** (★). A win lands in the shaded region: cheaper *and* no worse. Always-cheapest is cheapest on average, not per episode, which is why a few points sit left of it. Nested cross-validation; 10,000 bundle permutations.

The ceiling is real. In hindsight, choosing the eyes per task solves **+3.45 to +16.35 pp** more than the best single fixed mode, at 1.6–35.3% lower cost, in 8 of 8 settings.

Nothing we trained wins. 0 of 8 learned routers beat always-cheapest on both success and cost — and even the hindsight oracle does so in only **1 of 8**.

What survives is a bound, not a router. Sending the tasks nobody solves to the cheapest mode saves 9.5–30.6% at identical success in 8 of 8 — against the best-success fixed mode, and plain always-cheapest usually saves more.

WHY IT CANNOT BE LEARNED

A routing label exists only when the agent solves a task. Here the best single mode solves just **2–36%** of tasks, leaving typically **15–97** usable labels per setting — **the agents that would gain most from routing produce the least supervision to learn it**. Deliberately shrinking the training data confirms scarcity is the mechanism and prices it: the failing settings would need at least **2.1–4.2×** more tasks than the benchmarks contain.

Rerunning **one unchanged mode** flips **10–14%** of outcomes and by itself buys **2.0–7.6 pp** (B0 · classifieds, six replicated modes, $n=224$); every gain on this sheet is read against that band, not against zero.

TAKEAWAY

Routing is not only a model-selection problem: its learnability depends on the competence of the agent producing the labels. So, in this order: improve the agent, then generate reliable supervision, then learn selective perception.

Measured inside the 2–36% success regime we observed. This conclusion need not hold for stronger agents.

